

Role of Specialty Pharmacist in Cannabidiol Use

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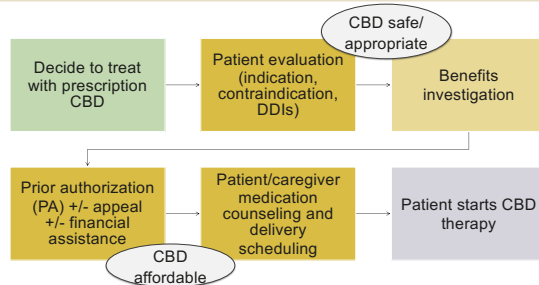
BACKGROUND

- Prescription cannabidiol (CBD) is approved for the treatment of Dravet, Lennox Gaustaut, and Tuberous Sclerosis Syndromes as adjunct therapy in combination with other anti-epileptic drugs (AEDs).¹
- CBD may affect multiple metabolism pathways, leading to a variety of pharmacodynamic (PD) and pharmacokinetic (PK) drug-drug interactions (DDI). Therefore, specialty pharmacists can play an important role in ensuring safe treatment initiation and management.

STUDY OBJECTIVE

Describe the number and type of actions performed by a neurology specialty pharmacist at time of prescription CBD initiation

Figure 1. Role of Integrated Neurology Specialty Pharmacist



Care Team Member



METHODS

- Design** Single-center, retrospective cohort study
- Inclusion** All patients prescribed CBD for the management of a seizure disorder by the center's outpatient neurology clinics from January 1, 2019 – April 30, 2020
- Exclusion** Access and fulfillment of prescription CBD was not handled by the integrated specialty pharmacy or participation in a prescription CBD clinical trial
- Data sources** Electronic health record and specialty pharmacy management system

- PK Interactions** An interaction involving metabolism pathways which effects the risk of side effects, toxicity, or therapeutic control
- PD Interactions** An interaction involving additive side effect risks without altering medication levels

RESULTS

Table 1. Cohort Demographics (N = 136)

	Pediatric (N=92) % (n)	Adult (N=44) % (n)
Age, years [median, (IQR)]	10 (5 – 14)	28 (21 – 44)
Gender, female	47 (43)	57 (25)
Race, white	84 (77)	86 (38)
Insurance type		
Medicaid/Medicare	73 (67)	78 (34)
Commercial	20 (18)	23 (10)
Tricare	8 (7)	–
Height, cm [median, (IQR)]	130 (102 – 147)	164 (153 – 173)
Weight, kg [median, (IQR)]	29 (17 – 38)	62 (49 – 76)
Lennox-Gastaut Syndrome IQR = Interquartile range	89 (82)	80 (35)

Figure 2. Number and Type of Pharmacist Actions at CBD Initiation

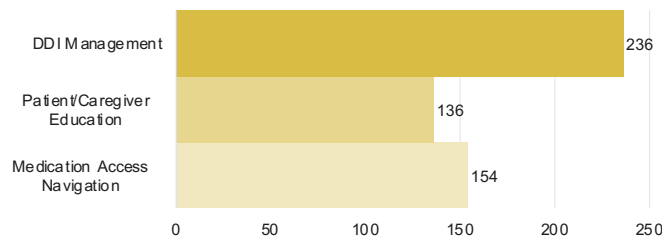
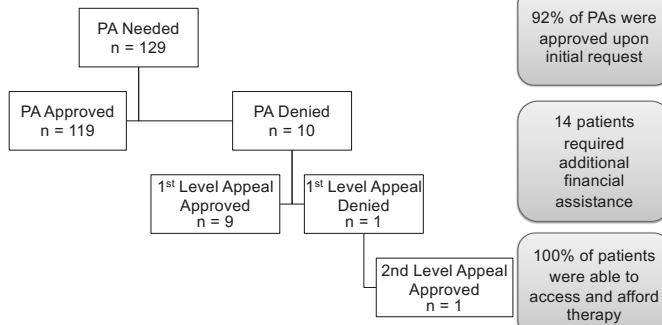


Figure 3. PA Outcomes



PA not required by 7 patients

Figure 4. Types of DDI

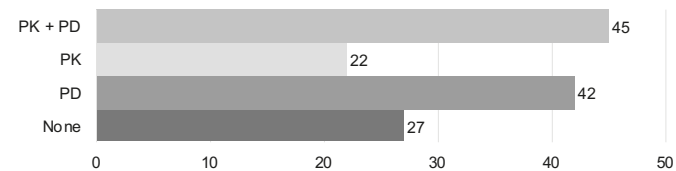
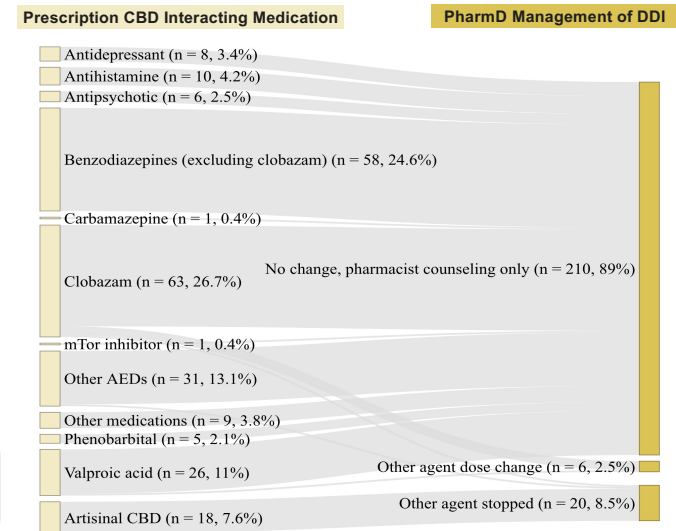


Figure 5. Outcomes of DDI Management (N = 236)



- The most common DDIs were benzodiazepines (24.6%), clobazam (26.7%), other AEDs (13.1%), and valproic acid (11%)

CONCLUSIONS

- Our study provides previously unavailable data on real-world management of patients utilizing an integrated specialty pharmacy at time of prescription CBD initiation.
- Our findings demonstrate the integral role of a neurology specialty pharmacist in securing insurance approval, educating patients, and managing possible DDIs to ensure safe initiation of prescription CBD therapy.
- Further research is ongoing to evaluate the role of a neurology specialty pharmacist in the long-term management of this patient population.